Fixed-charge gradients and the ordinary charged tail

CAR channel identities, marked Wick estimates and finite-time summation

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Abstract

We replace the full-Fock Riesz step in the ordinary charged-tail argument by estimates on adjacent charge sectors. Normalized up/down channels contract the principal row and column gradient terms; their remainders accumulate linearly along a channel path. A graded derivative shifts each body/size deficit by one half. This gives a direct marked-Wick gradient estimate with a square-root-p cost, uniform in polynomial degree. A rectangular completely positive map estimate controls products at fixed Schatten orders. The actual covariance residual and forced Jacobi recurrence then yield a finite-N, degree-uniform ordinary tail in a central charge window. Combined with the previous fixed-degree half-filling bridge, this permits finite-time summation in the previously controlled polynomial preparations.

1 Model, norms and statement

Use standard CAR, $\{c_i, c_j^\dagger\} = \delta_{ij}$, and $\mathcal{H}_m = \wedge^m \mathbb{C}^N$, $d_m = \binom{N}{m}$. The Hermitian Gaussian overlapping p-body Hamiltonian is the same common-disorder model as in the preceding drafts. Its directed coefficients are $A_{IJ} = \sqrt{w_N} C_I^\dagger C_J$, $|I| = |J| = p$, with covariance

$$C_m I = \mathbb{E} H_m^2, \qquad C_m = w_N \binom{m}{p}\binom{N-m+p}{p}.$$

Take even N, $\lambda_N = p^2/N \in [\lambda_-, \lambda_+] \subset (0, \infty)$, and fixed energy normalization $J_s > 0$. Along the double-scaling sequence p tends to infinity. Covariances are bounded above by C_* on all sectors and bounded below on $|m - N/2| \le L_0 p$ for each fixed L_0, by the binomial estimates in [4]. All norms include Gaussian expectation:

$$\|X\|_{s,m} = (\mathbb{E}\,\mathrm{tr}_m\,|X|^s)^{1/s}, \qquad \mathrm{tr}_m = d_m^{-1}\,\mathrm{Tr}_{\mathcal{H}_m}\,.$$

On neutral block families put

$$TX = \sum_i c_i^\dagger X c_i, \quad SX = \sum_i c_i X c_i^\dagger, \quad D = Q - T, \quad \bar{D} = N - Q - S, \quad L = (D + \bar{D})/2.$$

Loss acts on the full family before restriction. $\mathcal{N}$ is the coupling Gaussian number, not body degree; $Z_n = \mathcal{W}_n(H^n)$ is top coupling chaos. Fixed-charge ensembles have the established background of [2]; the present finite-N bounds concern this specific model.

Theorem 1. *For fixed finite s and L_0, constants K_s, L_s, b_s independent of N and n satisfy*

$$G_{n,N;m} = C_m^{-n/2}[p\mathcal{N} - D - \lambda_N \mathcal{N}(\mathcal{N}-1)]H^n\big|_{\mathcal{H}_m},$$
$$\sup_{|m-N/2|\le L_0 p} \|G_{n,N;m}\|_{s,m} \le K_s (n+1)^{b_s} L_s^n \sqrt{n!}. \tag{1}$$

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C_m here is a fixed target scalar applied after loss. For finite S, the actual series $G_N(t) = \sum_{n\geq 0}(it)^n G_{n,N;m}/n!$ has tail

$$\sup_N \sup_{|t|\leq S} \left\| G_N(t) - \sum_{n\leq K} (it)^n G_{n,N;m}/n! \right\|_{s,m} \leq K_s \sum_{n>K} \frac{(n+1)^{b_s}(L_s S)^n}{\sqrt{n!}} \longrightarrow 0. \tag{T}$$

Time uses dimensionless $h = H_m/\sqrt{C_m}$ units and $\hbar = 1$. Finitely many initial N are absorbed in the constants.

The fixed-degree half-filling bridge is in [6]. The full-Fock Clifford estimate of [1], Theorem 6.8, is not a fixed-charge theorem; we do not restrict it to a sector. We use the earlier pure-tag and covariance identities [3, 4, 5], with their sectorwise bounds, and supply the missing gradient and product estimates.

2 Adjacent-sector gradients and channel transport

Put $Y_{i,m}(X) = [c_i, X]_m : \mathcal{H}_m \to \mathcal{H}_{m-1}$ and

$$g_s^C(X;m) = \left\| (\sum_i Y_{i,m}^\dagger Y_{i,m})^{1/2} \right\|_{s,m}, \qquad g_s^R(X;m) = \left\| (\sum_i Y_{i,m} Y_{i,m}^\dagger)^{1/2} \right\|_{s,m-1}.$$

Let g_s be the maximum of both quantities for X and $X^\dagger$. Raising derivatives are covered by adjoints. Interval suprema include the neighboring sectors of their edges. We work at $s \geq 2$; higher even orders imply other fixed orders by monotonicity.

The channels $(tX)_k = (TX)_k/k$, $(sX)_k = (SX)_k/(N-k)$ are unital and preserve normalized sector trace, by cyclicity and $d_{k-1}/d_k = k/(N-k+1)$. Thus they contract every Schatten order, also after Gaussian averaging. The channel s and norm exponent s are distinguished by context.

Lemma 1. *On an interior edge $m \to m-1$, with $d = N - m + 1$,*

$$[c_i, tX]_m = -\frac{1}{m}\sum_j c_j^\dagger Y_{i,m-1}(X) c_j + \frac{1}{m}(X - tX)_{m-1} c_i, \tag{2}$$

$$[c_i, sX]_m = -\frac{1}{d}\sum_j c_j Y_{i,m+1}(X) c_j^\dagger + \frac{1}{d} c_i (sX - X)_m. \tag{3}$$

The principal terms contract both row and column square-function norms with factors $\sqrt{(m-1)/m}$ and $\sqrt{(d-1)/d}$.

Proof. CAR gives $[c_i, TX] = X c_i - \sum_j c_j^\dagger [c_i, X] c_j$ and $[c_i, SX] = -c_i X - \sum_j c_j [c_i, X] c_j^\dagger$. The left/right normalized denominators are m and $m-1$ for T, and $d-1$ and d for S. Keeping them distinct gives the identities.

For arbitrary $Y_i : \mathcal{H}_{m-1} \to \mathcal{H}_{m-2}$, let $P_i = \sum_j c_j^\dagger Y_i c_j$, $Q_C = \sum_i Y_i^\dagger Y_i$, $Q_R = \sum_i Y_i Y_i^\dagger$. Row/column operator Cauchy–Schwarz gives

$$\sum_i P_i^\dagger P_i \leq (m-1) T Q_C, \qquad \sum_i P_i P_i^\dagger \leq m T Q_R.$$

The first T targets m and the second targets $m-1$. After division by m^2, both yield $(m-1)/m$ times normalized t. For $Y_i : \mathcal{H}_{m+1} \to \mathcal{H}_m$ and $P_i = \sum_j c_j Y_i c_j^\dagger$,

$$\sum_i P_i^\dagger P_i \leq d S Q_C, \qquad \sum_i P_i P_i^\dagger \leq (d-1) S Q_R.$$

After division by d^2, both yield $(d-1)/d$ times normalized s. Positivity, Schatten monotonicity, and channel contraction at order $s/2$ prove the bounds. The proof also applies to $X^\dagger$. □

For the remainder $U_{m-1}c_i/m$, the column and row norms are, respectively, bounded by

$$m^{-1/2}\,\|U\|_{s,m-1}\,, \qquad \sqrt{N-m+1}\,\|U\|_{s,m-1}\,/m,$$

with equality in the row expression. For c_iV_m/d they are $\sqrt{m}\,\|V\|_{s,m}\,/d$ and at most $\|V\|_{s,m}\,/\sqrt{d}$. These follow by summing the Gram matrices before taking norms. On $[N/4, 3N/4]$, each step therefore adds at most $8N^{-1/2}\sup\|X\|_s$, since $U = X - tX$, $V = sX - X$. For a normalized path P of length ℓ, all of whose required edges and sectors stay in that interval, induction gives

$$g_s(PX;m) \le \sup_{\text{reachable}} g_s(X) + \frac{8\ell}{\sqrt{N}} \sup_{\text{reachable}} \|X\|_s\,. \tag{P}$$

Intermediate norms contract. No factor greater than one is raised to ℓ. For a scalar charge function f the extra derivative is $(f_m - f_{m-1})X_{m-1}c_i$, so

$$g_s(fX;m) \le \sup|f|\,g_s(X;m) + \sqrt{N}\sup|\Delta f|\sup\|X\|_s\,. \tag{F}$$

3 The graded shift and the pure marked gradient

For homogeneous parity $a \in \{0,1\}$ define algebraically

$$D_gX = \tfrac{1}{2}\{Q,X\} - (-1)^a\sum_i c_i^\dagger X c_i, \quad \partial_bX = bX - (-1)^a Xb, \quad L_g = D_g - \tfrac{1}{2}\sum_i \partial_{c_i^\dagger}\partial_{c_i}.$$

On neutral even fields these agree with D,L. No odd-field CP property is assumed. A normal-ordered monomial of total degree d is a D_g eigenvector with eigenvalue $d/2$. A graded derivative deletes one matching letter with its position sign. Such monomials form a basis; hence for $b = c_i$ or $c_i^\dagger$,

$$[\partial_b, D_g] = \tfrac{1}{2}\partial_b, \qquad [\partial_b, L_g] = \tfrac{1}{2}\partial_b. \tag{G}$$

For L, use the anticommutation of graded derivatives: their quadratic contraction commutes with each derivative. For an ordered list $T_j \in \{D, L\}$,

$$\partial_b \prod_{j=1}^{k}(np - T_j)Z_n = \prod_{j=1}^{k}(np - \tfrac{1}{2} - T_{j,g})\,\partial_bZ_n. \tag{S}$$

The order is preserved; commutativity of deficits is not required.

Lemma 2. *For fixed k,s and a fixed interior charge interval, uniformly in N,n,*

$$g_s\left(\prod_{j=1}^{k}(np - T_j)Z_n; m\right) \le K_{k,s}\sqrt{p}\,(n+1)^{2k+1}L_{k,s}^n\sqrt{n!}. \tag{W}$$

The field without the gradient has the earlier bound $K_{k,s}(n+1)^{2k}L_{k,s}^n\sqrt{n!}$.

Proof. We extend the restricted-covariance proof of [3]. The graded Leibniz rule for ∂_bZ_n chooses one of n original coefficient slots and deletes a matching letter, imposing one membership in its original I or J label. The remaining total degree is $2np - 1$, so (S) uses exactly the appropriate deficit. The graded product rule is

$$D_g(XY) = (D_gX)Y + X(D_gY) + (-1)^{|X|}\sum_i(\partial_{c_i^\dagger}X)(\partial_{c_i}Y).$$

Each subsequent deficit either multiplies a term by its number of already deleted pairs or creates a new two-letter contraction; L also permits an own-slot I/J contraction. After k operations there are at most $K_k(n+1)^{2k+1}$ terms, with coefficients depending only on k. Deleting an already deleted

letter is zero. All surviving memberships remain requirements on the original sets; (S) avoids differentiating inserted sandwich letters.

Expand a column square-function moment $\mathbb{E}\,\mathrm{tr}_m(\sum_i Y_i^\dagger Y_i)^r$ (order $2r$). There are $2r$ differentiated Wick blocks, at most $2rk$ pure tags, and r summed gradient indices, each occurring twice. Fix the Gaussian pairing, slot choices, and partition of all site variables into equal sites. Let A count distinct sites carrying a pure tag and B sites carrying gradients only. Every tagged site requires two distinct underlying p-set labels. For a cross-slot tag its two slots are in one Wick block and cannot be paired together; an own-slot tag uses independent I,J. Identifications elsewhere cannot remove those two requirements. Every gradient-only site needs at least one membership, and $B \le r$.

For fixed membership sets T_I, T_J, the occupation-basis column covariance is bounded by

$$C_* \frac{(p)_{|T_J|}}{(m)_{|T_J|}} \frac{(p)_{|T_I|}}{(N-m+p)_{|T_I|}} \le C_* (K_{k,r} p/N)^{|T_I|+|T_J|}. \tag{R}$$

If an I site is occupied but not removed by J the contribution is zero, so dropping that extra restriction is an upper bound. The row bound swaps I,J. Combine conditions at both ends of a Gaussian chord before applying (R). Derived coefficients are finite sums of fixed CAR sandwiches of the restricted original coefficient; their number depends only on k,r and all insertions contract.

Scan a chord word with column creation and row annihilation maps for Gaussian-index ancillas. Crossings only permute open ancillas unitarily. The contraction norm is bounded by the product of endpoint covariance square roots, keeping CAR signs throughout. Each independent set/site condition is counted once per chord. Summing site assignments therefore costs at most

$$N^{A+B}(p/N)^{2A+B} = \lambda_N^A p^B \le K_{k,r} p^r. \tag{C}$$

Additional memberships improve the estimate. The numbers of site partitions and intermediate charge shifts depend only on k,r, not n. For sufficiently large N all intermediate charges remain in a wider fixed interior interval. Small N have bounded p and CAR norms and are absorbed with polynomial degree costs.

There are at most $(2rn-1)!!$ Gaussian pairings, with $[(2rn-1)!!]^{1/(2r)} \le (2r)^{n/2}\sqrt{n!}$. Taking the root gives the stated slot factor and $\sqrt{p}$ from (C). The row moment has the same argument on sector $m-1$; adjoints and reversed orientations give the same estimates. Higher even orders imply all fixed orders. Omitting the gradient gives $B=0$ and the previous pure-tag estimate. □

4 Products without an ancilla dimension loss

For $\Phi(X) = \sum_i V_i X V_i^\dagger$ from an a-dimensional sector to a b-dimensional sector, the normalized-trace adjoint is $\Phi^\sharp(Y) = (a/b)\sum_i V_i^\dagger Y V_i$. For positive X,Y, spectral decomposition and scalar Holder with weights $\sum_i |(V_i)_{uv}|^2/b$ imply

$$\mathbb{E}\,\mathrm{tr}_b\, Y\Phi(X) \le (\mathbb{E}\,\mathrm{tr}_a\, X^u \Phi^\sharp(I))^{1/u} (\mathbb{E}\,\mathrm{tr}_b\, Y^v \Phi(I))^{1/v}, \qquad u^{-1}+v^{-1}=1.$$

The map and inputs may share the disorder. With $u = 2r$, duality at order r and a second trace Holder give

$$\|\Phi(X)\|_{r,b} \le 4\,\|X\|_{4r,a} \left\|\Phi^\sharp(I)\right\|_{2,a}^{1/(2r)} \|\Phi(I)\|_{2r-1,b}^{1-1/(2r)}. \tag{CP}$$

The factor four allows arbitrary X by splitting its real and imaginary positive/negative parts; positive X needs no factor. At $r=1$ the dual variable is bounded and the same formula holds.

Apply (CP) to the arrays $Y_i(X)$ and their adjoints. Adjacent dimension ratios are bounded in the interior. The ordinary Leibniz rule, with (CP) for inside Gram multipliers and rectangular Holder for outside multipliers, gives

$$\begin{aligned} g_s(XY;m) \le K_s\big[& g_{4s}(X;m) \sup_{k=m-1,m} \|Y_k\|_{4s,k} \\ & + \sup_{k=m-1,m} \|X_k\|_{4s,k}\, g_{4s}(Y;m)\big]. \end{aligned} \tag{B}$$

For example the row Gram of $Y_i(X)Y_m$ is $\Phi_X(Y_m Y_m^\dagger)$; its column Gram is $Y_m^\dagger(\sum_i Y_i(X)^\dagger Y_i(X))Y_m$. Use (CP) at $s/2$ for the former and Holder for the latter. The other derivative term and adjoints cover all cases. Thus no trace dimension of $I_N \otimes Y$ has been discarded.

For the body-loss cross term,

$$B_D(X,Y) = \sum_i [c_i^\dagger, X][Y, c_i], \qquad D(XY) = (DX)Y + X(DY) - B_D(X,Y),$$

rectangular Holder yields $\|B_D(X,Y)\|_{r,m} \le g_{2r}(X;m) g_{2r}(Y;m)$. The reverse loss uses row norms on edge $m+1$, and L is their average. No general fixed-charge Riesz theorem is used.

5 Residuals, ordinary powers and the forced recurrence

Use nested intervals $[2N/5, 3N/5] \subset [N/3, 2N/3] \subset [N/4, 3N/4]$. For large N, $p \le N/100$, leaving room for paths and a fixed number of neighboring sectors. Write $F_s(n) = K_s(n+1)^b L_s^n \sqrt{n!}$, allowing constants and b to increase a fixed number of times.

The covariance and scalar approximation are

$$\Phi_m(X) = w_N \sum_{I,J} C_I^\dagger C_J X C_J^\dagger C_I, \qquad \Psi_m = \Phi_m / C_m,$$

$$\theta_{v,m} = \prod_{j=0}^{p-1} (1 - vp/(m-j))_+ (1 - vp/(N-m+p-j))_+, \qquad A_v = p(\Phi - C\theta_v) Z_v.$$

Ψ is the exact normalized path of p down and p up channels [4]. If $vp \le N/8$, its telescoping identity is

$$A_v = pC \sum_{h=1}^{2p} f_h(Q) P_h E_h, \qquad E_h \in \{(vp - D)Z_v, (vp - \bar{D})Z_v\}. \tag{A}$$

Each P_h has length at most $2p$; f_h is one inverse charge denominator times remaining scalar factors, with exact shifts. The sectorwise bounds in [5] are

$$|f_h| \le K/N, \quad |\Delta f_h| \le K(1 + v\lambda_N)/N^2, \quad C \le C_*, \quad |\Delta C| \le Kp/N.$$

Lemma 2, (P) and (F) give

$$g_s(A_v)/\sqrt{p} \le F_s(v). \tag{A1}$$

Indeed the principal coefficient sum is $O(p^2/N)$ and the path cost is $O(p/\sqrt{N})$ before division by $\sqrt{p}$. After that division the scalar-gradient contribution is at most $Kp^{3/2}[p + 1 + v\lambda_N]/N^{3/2}$ times the pure envelope, with only polynomial v growth. If $vp > N/8$, use the coarse $g_s(X) \le K\sqrt{N} \sup \|X\|_s$ and $\|A_v\|_s \le Kp \|Z_v\|_s$. Since $p \le 8\lambda_N v$, $N \le 64\lambda_N v^2$, its cost divided by $\sqrt{p}$ is also polynomial in v.

The previous sectorwise derivative calculation gives

$$\|A_v\|_s + \|DA_v/p\|_s + \|LA_v/p\|_s \le F_s(v). \tag{A2}$$

Its inputs, which did not use Riesz, are the exact relations

$$[D,t] = t - t^2, \quad [D,s] = -s + \frac{Q}{N-Q} ts - \frac{2Q-N}{N-Q} I, \quad [D,f] = (f(Q) - f(Q-1))T$$

and reverse-loss versions. Expanding a path gives $O(\ell/p)$ for $[D/p, P]$ and $[L/p, P]$. Repeated pure deficits control derivatives of E_h in (A); the scalar differences control the remaining terms. The large-v case uses the same $N = O(v^2)$ estimate.

All-sector ordinary Gaussian moments obey $\|H^w\|_s \le L_s^w \sqrt{w!}$. Lemma 2 at $n=1, k=0$ gives both force Grams of $[c_i, H]$ of order p. Hence (CP) bounds each forward/reverse force map by $K_r p \|X\|_{4r}$. Expanding

$$[c_i, H^w] = \sum_{a+b=w-1} H^a_{m-1} [c_i, H]_m H^b_m$$

and applying this force estimate to inside Gram multipliers gives $g_s(H^w)/\sqrt{p} \le F_s(w)$ at fixed higher norm orders. For D the exact expansion is

$$DH^w = pwH^w - \sum_{a+b+c=w-2} H^a \Gamma_H(H^b) H^c, \qquad \Gamma_H(X) = \sum_i [c_i^\dagger, H] X [H, c_i].$$

For L use single-slot LH and averaged forward/reverse cross terms. Pure $n=1$ controls LH/p. Fixed-order Holder and (CP) then give

$$\|H^w\|_s + \|DH^w/p\|_s + \|LH^w/p\|_s + g_s(H^w)/\sqrt{p} \le F_s(w). \tag{H}$$

Only $O(w^2)$ terms occur, each with at most three powers and one force map. Norm orders are not doubled w times.

Define the finite-N Jacobi polynomials and actual error by

$$Q_{j+1,m}(x) = xQ_{j,m}(x) - b_{j,m} Q_{j-1,m}(x), \qquad b_{j,m} = C_m \sum_{v<j} \theta_{v,m},$$
$$\eta_j = p[Z_j - Q_j(H)], \qquad Q_0 = 1, \qquad Q_1 = x.$$

The exact forced recurrence [5] is

$$pr_\ell = \sum_{v+w=\ell-1} \mathcal{W}_{v+w}(A_v H^w), \qquad \eta_j = -\sum_{\ell<j} P_{j,\ell,m}(H)\, pr_\ell. \tag{J}$$

$P_{j,\ell}$ is the scalar Jacobi propagator, of degree at most $j-\ell-1$. Its coefficient sums grow at most exponentially, since $b_{j,m} \le B$ on interior intervals. Their charge differences satisfy $N \sum |\Delta f|/p \le \mathrm{poly}(j) K^j$, from $|\Delta b_{j,m}| \le K[p+j^2]/N$ and path product differences.

The OU/Chebyshev estimate $\|\mathcal{W}_d X\|_s \le 2^d \|X\|_s$ for degree-d Gaussian polynomials [4] applies unchanged to row/column arrays: conditional Jensen uses their Banach norms with the specified sector trace. Physical derivatives commute with coupling OU and top-chaos projection.

Apply (A1), (A2), (H), (B) and the loss cross bound first to $A_v H^w$, then top project and sum. Fully expand $P_{j,\ell}(H)$ and apply the same estimates to $H^a(pr_\ell)$, retaining coefficient differences. There are two product layers, independent of j. Norm orders remain fixed multiples of s. Use $\sqrt{a!v!w!} \le \sqrt{j!}$, exponential coefficient sums and polynomially many degree choices to obtain

$$\|\eta_j\|_s + \|D\eta_j/p\|_s + \|L\eta_j/p\|_s + g_s(\eta_j)/\sqrt{p} \le F_s(j). \tag{E}$$

For coefficient gradients (F) uses $\sqrt{N}|\Delta f|/\sqrt{p} \le \sqrt{p/N}\,\mathrm{poly}(j)K^j$; for D/L it uses $N|\Delta f|/p$. All needed neighbors fit the nested intervals; no sharp charge cutoff is differentiated.

6 Ordinary powers and selected time fields

Write $H^n = \sum_j h_{nj,m} Q_{j,m}(H)$ using nonnegative weighted Jacobi paths. The exact covariance ratio and the Lipschitz bound for the positive part in θ_v give [5]

$$\sum_j |h_{nj,m}| \le K^n, \qquad \sum_j N|\Delta h_{nj,m}| \le K(n+1)^3 L^n (1+|X_N|), \quad X_N = (m - N/2)/p. \tag{I}$$

Explicitly,

$$\frac{C_m}{C_{m-1}} - 1 = \frac{p(N-2m+p+1)}{(m-p)(N-m+p+1)}, \qquad N|\Delta\theta_{v,m}| \le K\lambda_N v.$$

The second bound includes positive-part cutoffs. The factor $1 + |X_N|$ is bounded on our central window; it cannot be discarded on a macroscopic charge interval.

Keeping every charge difference yields

$$(p\mathcal{N} - D)H^n = \sum_j [h_{nj}(pj - D)Z_j - (\Delta h_{nj})TZ_j - h_{nj}\mathcal{N}\eta_j + h_{nj}D\eta_j/p + (\Delta h_{nj})T\eta_j/p]. \tag{O}$$

Pure deficit bounds, (I) and normalized T control the first two terms. The third uses the degree-j Gaussian number multiplier and (E), with Chebyshev cost at most jK^j. The last two use (E), (I) and adjacent-sector contraction. The counterterm $\lambda_N\mathcal{N}(\mathcal{N}-1)H^n$ has the same multiplier estimate with an extra polynomial in n. Thus (1) holds before the target factor $C_m^{-n/2}$; its lower bound changes only L_s. Finitely many small N have bounded p and physical losses, hence the same Gaussian factorial envelope. Summing proves (T), also for fixed time derivatives and fixed exterior energy powers. This completes Theorem 1.

At $m = N/2$, combine (T) with the same-copy fixed-degree bridge [6]. Exact energy pinching contracts every fixed sector Schatten norm. Each previously controlled fixed polynomial preparation P has a finite-N orbit density satisfying $\sup_N \|R_{P,N}\|_a < \infty$ at each fixed a [7]. Holder transfers the actual tail to that preparation, with the degree of P fixed. Fixed products in a covariant block use a fixed number of Holder factors. Take N to infinity at fixed truncation, then remove the truncation with (T). This gives the complete ordinary charged time series and its exactly pinched version in that preparation class.

The complete selected bilocal still requires a sectorwise audit of the actual neutral/rank filter and memory-return estimates. Spatial/mass generators and finite-G dynamics remain open. The earlier full-trace bilocal construction is neither retracted nor promoted to a selected-state result by this theorem.

7 Finite checks and provenance

The saved channel check uses $N = 3, 4, 5, 6$, three non-selfadjoint integer neutral fixtures per N and their transposes. It verifies 1,032 gradient and 216 unital/normalized-trace identities exactly after clearing denominators. A further 168 PSD and 324 Schatten checks are floating-point diagnostics, not proofs of the lemma or its path corollary.

The new graded check exhausts 340 matrix units and 340 normal monomials at $N = 1, 2, 3, 4$. It verifies 5,008 derivative commutators, 840 repeated-deficit shifts, 340 Euler identities and 2,504 signed deletions, 1,252 of them nonzero. Doubled maps use exact integer arithmetic. These fixtures check algebra and signs, not degree-uniform Gaussian convergence. The general estimates rely on the contraction and norm arguments above, not numerical extrapolation. Successful scientific outputs are preserved without rerunning their drivers.

References

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